Edible vaccines are prepared by molecular pharming using the science of genetic engineering. Selected genes are introduced into plants and the transgenic plants are induced to manufacture the encoded protein. Edible vaccines are mucosal targeted vaccines which cause stimulation of both systemic and mucosal immune response. At present edible vaccines are produced for human and animal diseases like measles, cholera, foot and mouth disease and hepatitis.

DNA Vaccines

Genetic immunisation by using DNA vaccines is a novel approach that came into being in 1990. The immune response of the body is stimulated by a DNA molecule. A DNA vaccine consists of a gene encoding an antigenic protein, inserted onto a plasmid, and then incorporated into the cells in a target animal. DNA instructs the cells to make antigenic molecules which are displayed on its surfaces. This would evoke an antibody response to the free floating antigen secreted by the cells. The DNA vaccine cannot cause the disease as it contains only copies of a few of its genes. DNA vaccines are relatively easy and inexpensive to design and produce.

Vaccines produced by these new techniques have definite advantages like producing target proteins, long lasting immunity and trigger immune response only against specific pathogens with less toxic effects.

Recombinant hepatitis B vaccine as a subunit vaccine is produced by cloning hepatitis B surface antigen (HbsAg) gene in the yeast, *Saccharomyces cerevisiae* **(Fig. 10.2)**.

The recombinant vaccine for hepatitis B (HbsAg) was the first synthetic vaccine launched in 1997 which was marketed by trade names Recombivax and Engerix-B. India is the fourth country in the world after USA, France and Belgium to develop an indigenous hepatitis B vaccine.

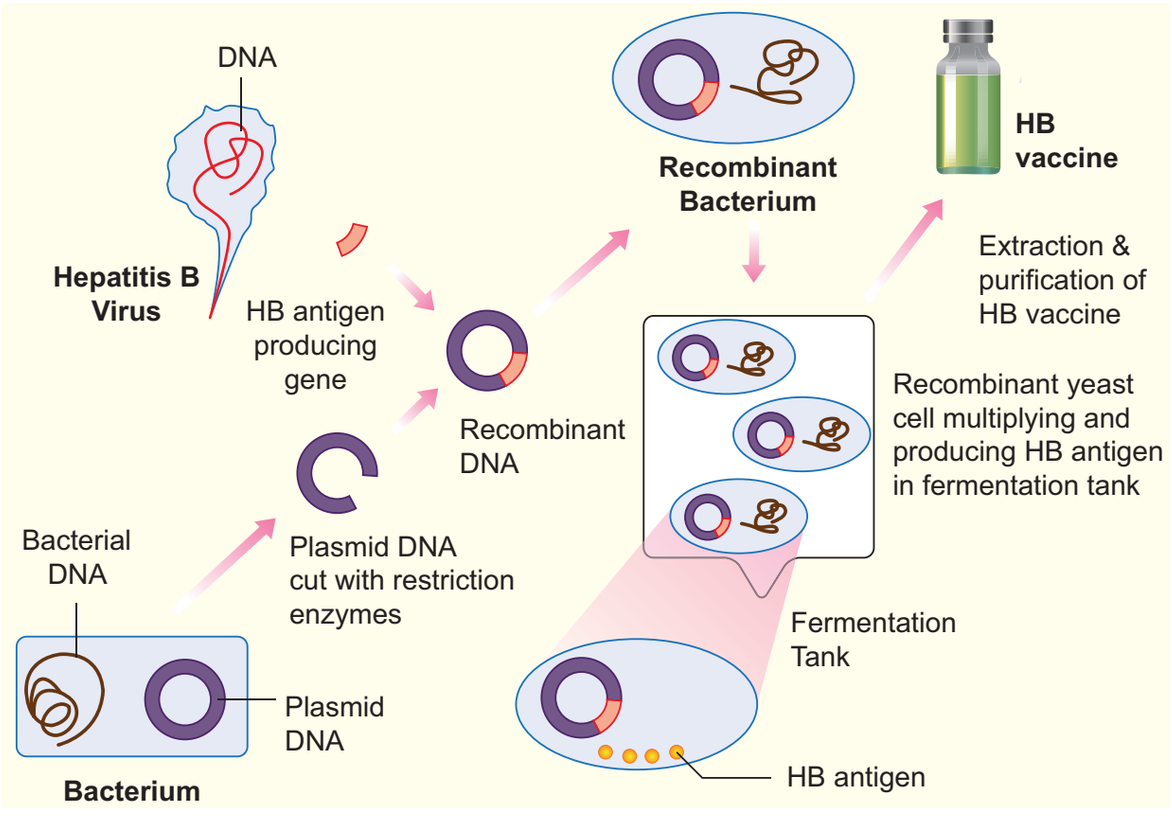


Fig. 10.2 Production of recombinant HB Vaccine

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